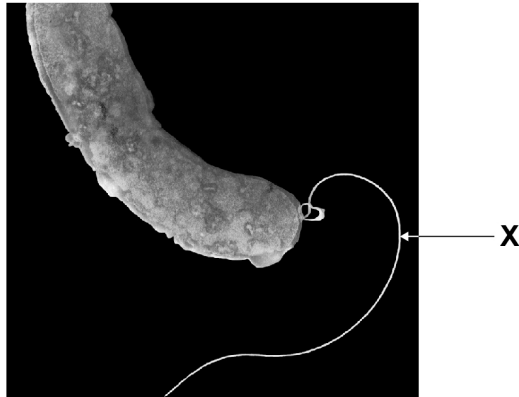


0 3 . 1

Figure 2 is an image of a bacterium obtained using a scanning electron microscope.

Figure 2



Name the structure labelled **X**.

[1 mark]

0 3 . 2

Figure 2 is different from an image of this bacterium obtained using a transmission electron microscope.

Describe and explain **one** difference between these images.

[2 marks]

Description _____

Explanation _____

0 3 . 3

The resolution of an image obtained using an electron microscope is higher than the resolution of an image obtained using an optical microscope.

Explain why.

[1 mark]

Question 3 continues on the next page

Turn over ►



0 3 . 4

A student determined the size of a cell structure from a photograph obtained using a microscope.

He used a ruler and a calculator and gave the answer in μm

Describe how the student determined the size of the structure.

[2 marks]

0 3 . 5

Name **two** structures found in **all** bacteria that are **not** found in plant cells.

[2 marks]

1

2

0 3 . 6

Name **two** features of HIV particles that are **not** found in bacteria.

Do **not** include attachment protein in your answer.

[2 marks]

1

2

10

